

Cohort Retention Analysis

Monthly retention of customers by first-purchase cohort | Online Retail II structure | Dec 2009 - Dec 2011

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5,877

Customers analysed

29.0%

Month-1 retention

24.4%

Month-6 retention

20.2%

Month-12 retention

1. Executive Summary

This report builds a monthly cohort retention analysis to understand how long customers keep buying after their first purchase. Customers are grouped by the month of their first order (their cohort), and we track the share of each cohort that places at least one order in each following month.

Headline result: only about 29% of customers return in the month after their first purchase, so roughly 71% buy once and leave. After that first drop the curve flattens and settles near 20%, a loyal core that keeps buying for the full two years. Repeat buying is also strongly seasonal, with clear peaks around September.

2. Objective, Data and Method

Objective: build a cohort retention table by signup month and understand retention over time.

Dataset: transaction data with the Online Retail II schema (Invoice, StockCode, Description, Quantity, InvoiceDate, Price, Customer ID, Country) covering Dec 2009 to Dec 2011. The raw file has 341,544 rows.

Note: the dataset used here is simulated to mirror the structure and data-quality issues of the real UCI dataset.

Data cleaning steps:

- Removed exact duplicate rows.
- Dropped rows with missing Customer ID (a cohort needs an identifiable customer).
- Removed cancelled invoices (Invoice starting with 'C').
- Kept only rows with Quantity > 0 and Price > 0.
- Excluded non-product codes (POST, DOT).

After cleaning, **261,431** rows (76.5%) remained, covering **5,877** customers, 38,880 invoices and total revenue of 15,418,316.

Cohort logic: CohortMonth = month of a customer's first purchase. CohortIndex = number of months between a purchase and the customer's cohort month (0 = first month). Retention % = customers active in month N divided by cohort size (month 0).

Validation: the same retention table was built independently in SQL (SQLite) and compared with the Python (pandas) result. The maximum difference was 0.0, so both methods agree exactly.

3. Cohort Retention Table (Summary)

Selected milestones for every cohort. Blank cells mean the cohort is too recent to have reached that month.
The full month-by-month table is shown as a heatmap in the next section.

Cohort	Size	M1 %	M3 %	M6 %	M12 %	M18 %
2009-12	618	25.7	24.1	20.6	16.0	15.7
2010-01	356	25.3	22.5	23.9	21.3	14.6
2010-02	348	29.0	24.1	24.1	21.6	19.3
2010-03	222	25.7	19.8	29.3	20.3	27.5
2010-04	216	30.1	23.6	26.9	19.4	18.1
2010-05	220	23.6	20.0	24.1	17.3	20.0
2010-06	195	29.7	42.1	20.0	20.0	14.9
2010-07	194	29.9	31.4	23.2	20.1	
2010-08	210	37.6	37.1	18.6	12.9	
2010-09	215	35.3	25.6	25.6	27.9	
2010-10	205	36.6	23.9	21.5	22.4	
2010-11	220	23.2	22.3	21.8	22.3	
2010-12	206	28.6	19.9	23.3	20.9	
2011-01	197	27.4	26.9	23.9		
2011-02	203	18.2	19.2	21.2		
2011-03	218	25.7	26.1	29.8		
2011-04	180	22.8	20.0	31.1		
2011-05	201	25.4	24.4	31.3		
2011-06	211	27.0	31.8	22.7		
2011-07	209	25.4	39.7			
2011-08	212	42.5	40.6			
2011-09	218	40.4	33.9			
2011-10	199	36.7				
2011-11	204	24.0				
2011-12	200					
Average	235	29.0	27.2	24.4	20.2	18.6

4. Retention Heatmap

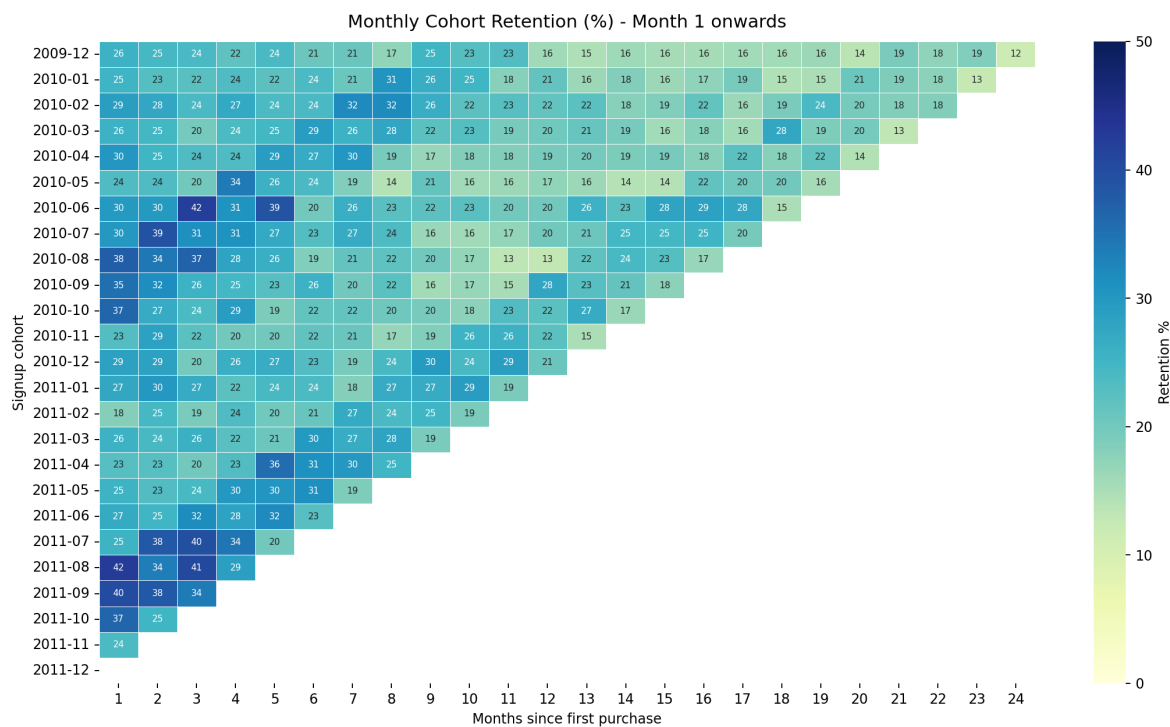


Figure 1: Retention % by cohort (rows) and months since first purchase (columns). Month 0 (always 100%) is omitted so the differences stay visible. Colour scale is capped at 50%.

Read the heatmap along diagonals: warm cells that line up diagonally fall in the same calendar month, which is the signature of seasonality rather than cohort quality.

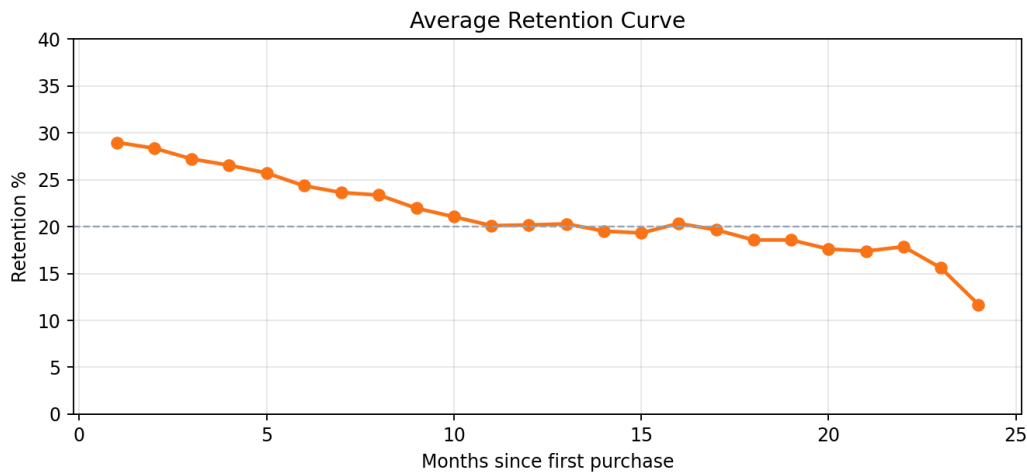


Figure 2: Average retention across all cohorts. The dashed line marks 20%.

5. Key Findings

1. The biggest drop happens in the first month.

Only 29.0% of customers return after their first purchase (about 71% never come back). Afterwards retention falls slowly: Month 3 = 27.2%, Month 6 = 24.4%, Month 12 = 20.2%. The curve flattens near 20%, which points to a loyal core.

2. Seasonality strongly influences repeat purchases.

Warm diagonals appear around September. The 2010-06 cohort peaks at Month 3 (42%), the 2010-07 cohort at Month 2 (39%) and the 2011-04 cohort at Month 5 (36%). All three fall in September, so repeat buying is driven by the pre-holiday season, not only by cohort quality.

3. Cohorts acquired in Aug-Sep retain best.

Best Month-1 retention: 2011-08 (42.5%), 2011-09 (40.4%) and 2010-08 (37.6%). Customers who join just before or during peak season are more likely to return.

4. Off-season cohorts churn faster.

Weakest Month-1 retention: 2011-02 (18.2%), 2011-04 (22.8%) and 2010-11 (23.2%). Customers acquired in slow months tend to leave sooner.

5. The earliest cohorts are the largest and most loyal.

The three largest cohorts are 2009-12 (618 customers), 2010-01 (356) and 2010-02 (348). Even after 24 months, the 2009-12 cohort still has roughly 12-19% of customers active.

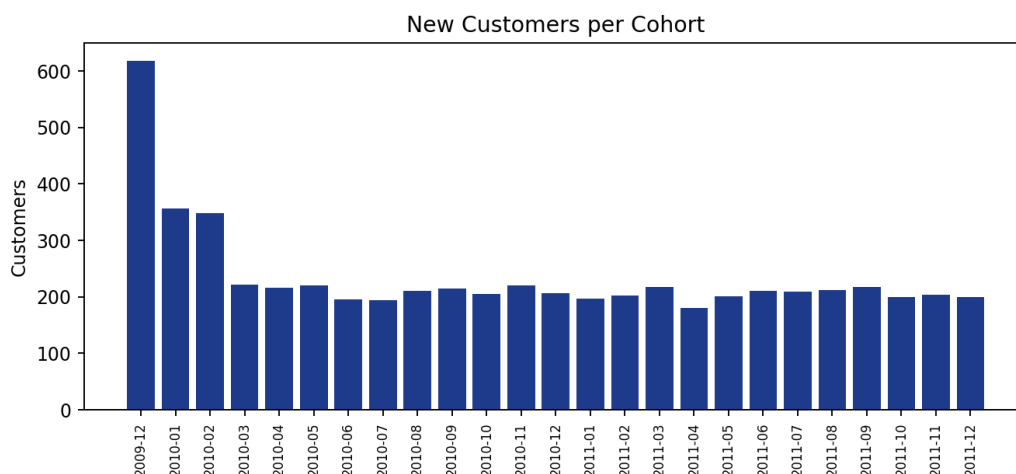


Figure 3: Number of new customers in each monthly cohort.

6. Recommendations

- **Improve onboarding:** send a second-purchase incentive (email or coupon) within 30 days of the first order, because this is where the largest loss occurs.
- **Time acquisition campaigns:** increase marketing spend in Aug-Sep, when new customers show the highest retention.
- **Support off-season cohorts:** run win-back or engagement campaigns for customers acquired in slow months such as February and April.
- **Reward the loyal core:** the roughly 20% of customers who stay long term likely drive a large share of revenue, so a loyalty programme is worth considering.

7. Limitations

- Recent cohorts (from 2011-10 onwards) have incomplete data and should not be compared directly with older cohorts.
- Retention is defined as 'any purchase in the month'. Revenue-based retention could tell a different story.
- The analysis uses a simulated dataset modelled on Online Retail II, so exact figures would differ on the real data.

8. Conclusion

Customer retention is low in the first month but stabilises at around 20%, and repeat buying is heavily seasonal. The business should focus on converting first-time buyers into second-time buyers early, and align acquisition and retention efforts with the seasonal peak to maximise long-term customer value.